Data Profiling - Observations & Gaps Register

Scope: all CSV files under __**initial_load** (5 domains, 13 datasets, 266 files). Profiling was done locally with Python on 2026-11-13; the files have not yet been loaded into Snowflake.

OBS = observation (a verified fact about the data; usually acceptable or informational). **GAP** = a deficiency that needs a decision or fix before the data can be trusted for that use. Severity: Critical > High > Medium > Low > Info.

Summary

Domain	Datasets	OBS	GAP	Critical	High	Medium	Low
Country & Reference	region, country, tax, currency (4 files)	6	7	0	3	3	1
Product	category, family, model, sku, availability (5 files)	7	6	0	2	2	2
Store	store_master (1 file)	6	6	0	2	3	1
Customer	customer_master (245 files)	7	9	0	2	4	3
Sales	sales_header, sales_item (14 files)	8	9	1	3	3	2
Total	13 datasets	34	37	1	12	15	9

Top priorities

GAP-001 (High, Country & Reference) - All 35 tax rows start on 2020-01-01 while sales start on 2019-01-01, so 2019 sales have no valid tax rate in an as-of join. Rates are current values (FI 25.5% from 2024, CH 8.1% from 2024, SG 9% from 2024, TR 20% from 2023), not historical ones.

GAP-006 (High, Country & Reference) - Effective tax rate in sales differs from tax_master for 5 countries: CA 13% vs 5%, BR 12% vs 17%, CH 7.7% vs 8.1%, MY 6% vs 10%, FI 24% vs 25.5% (45,671 transactions).

GAP-007 (High, Country & Reference) - tax_inclusive_flag = Y for 33 countries, but every sale computes net_total = gross - discount + tax (tax added on top), i.e. 451,922 contradicting transactions.

GAP-010 (High, Product) - No lifecycle variation: all families, models, SKUs and availability rows are ACTIVE/Y and every discontinue date is empty (e.g. iPhone 11 is still available in 35 countries).

GAP-011 (High, Product) - Availability is a full cross join (650 SKUs x 35 countries), including Vision Pro everywhere. It carries no real sell-ability signal.

GAP-020 (High, Store) - 152,808 POS transactions in 83 stores occur before store_open_date. Every store's first sale is 2019-01-01..06 even for stores opening in 2025-2026 (18 stores).

GAP-021 (High, Store) - All 70 tax_jurisdiction_code values (121 stores) are absent from tax_master, which only has country-level codes.

GAP-030 (High, Customer) - Loading all snapshot files into one table without snapshot_year yields 2.5 rows per customer and inflates customer counts ~2.5x.

GAP-031 (High, Customer) - 111,385 transactions occur before the customer's registration_date.

GAP-040 (Critical, Sales) - Amounts are USD-scale but labelled with local currencies (iPhone 11 sells for ~700-1,600 in USD, JPY, KRW, INR alike). 66,607 JPY/KRW transactions have fractional amounts although minor_unit = 0.

GAP-041 (High, Sales) - Tax is added on top in 33 tax-inclusive countries (451,922 transactions). See GAP-007.

GAP-042 (High, Sales) - 45,671 transactions use a rate different from tax_master (CA, BR, CH, MY, FI); rates are constant over 2019-2025 in sales, so there is no rate history anywhere.

GAP-043 (High, Sales) - 6,737 sales occur before the SKU global launch date (6,739 before local launch).

Country & Reference

ID	Type	Severity	File(s)	Observation / Gap	Recommendation
OBS-001	OBS	Info	region/country/tax/currency_master	All four files are structurally clean: UTF-8, LF endings, no nulls, no leading/trailing spaces, no ragged or duplicate rows. currency_master contains non-ASCII symbols (e.g. EUR, INR, AED, SAR symbols).	Load with ENCODING='UTF8'.
OBS-002	OBS	Info	country_master	country_master is the hub: 35 countries -> 5 regions, 27 currencies (EUR shared by 9 countries), 35 tax codes (strict 1:1, tax_code prefix = country_code). Zero orphans and zero unused reference rows.	Declare PK/FK constraints in DDL for documentation.
OBS-003	OBS	Low	country_master	apple_fiscal_segment maps 1:1 to region_code (AMER=Americas, EMEA=Europe, APAC=Rest of Asia Pacific, JAPAN=Japan, GREATER_CHINA=Greater China). It is redundant.	Keep as display label only.
OBS-004	OBS	Info	country_master	gdpr_applicable = Y for 15 countries (EU/EEA + UK + CH); consistent with regulation. ecommerce_supported = Y for all 35; retail_store_supported = Y for 24, N for 11.	None.
OBS-005	OBS	Low	all reference masters	Every reference row is active and open-ended (effective_end_date = 9999-12-31); no SCD history exists in the initial load.	Design merge logic for SCD2 going forward.
OBS-006	OBS	Info	all reference masters	created_at has microsecond precision and reflects the extract date 2026-04-17.	Use TIMESTAMP_NTZ(6).
GAP-001	GAP	High	tax_master	All 35 tax rows start on 2020-01-01 while sales start on 2019-01-01, so 2019 sales have no valid tax rate in an as-of join. Rates are current values (FI 25.5% from 2024, CH 8.1% from 2024, SG 9% from 2024, TR 20% from 2023), not historical ones.	Build a tax rate history (SCD2) with real effective dates.
GAP-002	GAP	Medium	country_master	AE and SA are mapped to APAC / Rest of Asia Pacific, while other Middle East and Africa markets (IL, TR, ZA) are in EMEA.	Confirm the business region mapping.
GAP-003	GAP	Medium	country_master	Country code 'UK' is not ISO 3166-1 alpha-2 (correct code is 'GB'). iso_alpha3 'GBR' is correct.	Map UK->GB or add an iso_alpha2 column for external joins.
GAP-004	GAP	Medium	tax_master	Flat single rates for multi-level tax systems: US 7% sales tax (real rates vary by state), CA GST 5% only (ignores HST/PST), BR ICMS 17% (varies by state).	Model sub-national jurisdictions (see GAP-021).
GAP-005	GAP	Low	currency_master	Currency symbols are ambiguous: '$' is used by 7 currencies, 'kr' and the yen symbol by several.	Never use the symbol as a key; display it with the currency code.
GAP-006	GAP	High	tax_master vs sales_header	Effective tax rate in sales differs from tax_master for 5 countries: CA 13% vs 5%, BR 12% vs 17%, CH 7.7% vs 8.1%, MY 6% vs 10%, FI 24% vs 25.5% (45,671 transactions).	Correct tax_master or add history; re-validate.
GAP-007	GAP	High	tax_master vs sales_header	tax_inclusive_flag = Y for 33 countries, but every sale computes net_total = gross - discount + tax (tax added on top), i.e. 451,922 contradicting transactions.	Decide which side is right; this defines how revenue is reported.

Product

ID	Type	Severity	File(s)	Observation / Gap	Recommendation
OBS-010	OBS	Info	product_* (5 files)	Clean hierarchy: category (10) -> family (43) -> model (111) -> SKU (650) -> country availability (22,750). Zero orphans and zero childless parents at every level.	Declare PK/FKs.
OBS-011	OBS	Info	family/model/sku	Code patterns hold 100%: family_code = category_code + launch_year; model_code = family_code + 2-digit sequence; sku_code = 'M' + 6 chars + 'LL/A' (US part-number format).	Enforce with validation rules DQ-027/028/038.
OBS-012	OBS	Info	product_category_master	reporting_segment has 4 values: iPhone, iPad, Mac (Mac + Displays), Wearables, Home and Accessories (WCH, AIR, TV, HMP, ACC, VPR).	Use for segment reporting.
OBS-013	OBS	Info	category/family	Families cover launch years 2019-2024 only. Category effective_start_date is the product-line launch (Mac 1984 ... Vision Pro 2024) and predates the catalogue.	Do not use category dates as availability dates.
OBS-014	OBS	Info	sku/availability	SKU global_launch_date always equals model launch_date; local launch is 0-32 days (avg ~7) after global launch.	None.
OBS-015	OBS	Medium	product_country_availability	local_part_number is shared across country groups (UK/IE, DE/CH/AT, AU/NZ, AE/SA) - 2,600 shared values - so it is not unique by itself. Unique key is (sku_code, country_code).	Do not use local_part_number alone as a key.
OBS-016	OBS	Info	product_sku_master	price_tier: Premium 306, Ultra 255, Standard 89. 8 models have SKUs in more than one tier.	None.
GAP-010	GAP	High	family/model/sku/availability	No lifecycle variation: all families, models, SKUs and availability rows are ACTIVE/Y and every discontinue date is empty (e.g. iPhone 11 is still available in 35 countries).	Source real discontinue dates; current-catalogue logic cannot be built.
GAP-011	GAP	High	product_country_availability	Availability is a full cross join (650 SKUs x 35 countries), including Vision Pro everywhere. It carries no real sell-ability signal.	Source real availability before using it to validate sales.
GAP-012	GAP	Medium	product_model_master	model_name 'Apple Watch Ultra 2' appears twice (WCH202302 for 2023, WCH202402 for 2024).	Use model_code as key; disambiguate the name.
GAP-013	GAP	Medium	product_sku_master	variant packs several attributes into one string with inconsistent separators ('256GB / Silver', '8GB/256GB SSD', 'Standard', 'Yellow').	Split into storage / memory / colour / size columns.
GAP-014	GAP	Low	product_sku_master	sku_code is effectively the US part number; sales use sku_code (confirmed), not local_part_number.	Document sku_code as global key.
GAP-015	GAP	Low	product_*	SKU stems and launch lags look synthetic (uniform distribution).	Do not draw business conclusions from launch lag.

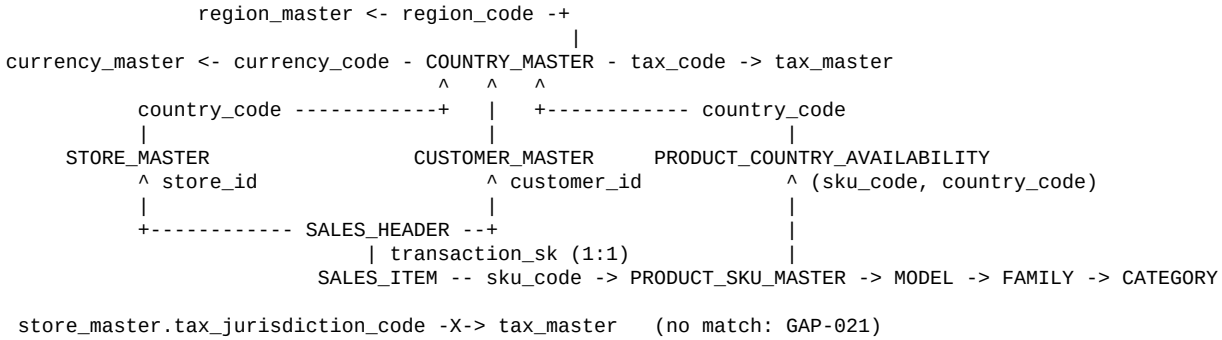
Store

ID	Type	Severity	File(s)	Observation / Gap	Recommendation
OBS-020	OBS	Info	store_master	121 stores in 24 countries - exactly the countries flagged retail_store_supported = Y. US 50, CN 10, UK 8, CA 6, AU 6, ...	None.
OBS-021	OBS	Info	store_master	format_code: MINI 43, FLG 40, MALL 38. Regions: AMER 58, EMEA 31, GREATER_CHINA 14, APAC 14, JAPAN 4.	None.
OBS-022	OBS	Low	store_master	region_code always equals the country's region_code - redundant.	Keep but validate (DQ-043).
OBS-023	OBS	Info	store_master	For US stores tax_jurisdiction_code always equals 'US_<state_code>_STD' (50/50).	None.
OBS-024	OBS	Info	store_master vs sales_header	Every store has POS sales in every year 2019-2025 (1,498 to 13,485 transactions per store, median 2,641). No store is unused and no sales reference an unknown store.	None.
OBS-025	OBS	Low	store_master	5 UK addresses contain embedded newlines inside quoted fields (file has 126 physical lines for 121 records). Names/cities/addresses contain CJK characters.	Use FIELD_OPTIONALLY_ENCLOSED_BY="" and UTF-8.
GAP-020	GAP	High	store_master vs sales_header	152,808 POS transactions in 83 stores occur before store_open_date. Every store's first sale is 2019-01-01..06 even for stores opening in 2025-2026 (18 stores).	Fix open dates or define effective open = first sale; flag in model.
GAP-021	GAP	High	store_master vs tax_master	All 70 tax_jurisdiction_code values (121 stores) are absent from tax_master, which only has country-level codes.	Create a tax_jurisdiction master with parent country tax_code.
GAP-022	GAP	Medium	store_master	Coordinates do not match addresses (e.g. US_0001 'VA' at lon -84.3; US_0004 'ND' at 28.7,-75.9 in the Atlantic). Cities/addresses are synthetic.	Do not use for geospatial analysis.
GAP-023	GAP	Medium	store_master	Rent per sq ft ranges from ~31 to ~3,690 USD/yr (29 stores > 1,000). Floor area does not vary by format (MINI up to 24,484 sq ft).	Treat store cost / productivity KPIs as unreliable.
GAP-024	GAP	Medium	store_master	effective_start_date = 2026-04-17 (load date) for all stores, so as-of joins from 2019-2025 sales find no version.	Set initial SCD2 version start to store_open_date or 1900-01-01.
GAP-025	GAP	Low	store_master	No closures: all stores ACTIVE, store_close_date empty.	None for now.

Customer

ID	Type	Severity	File(s)	Observation / Gap	Recommendation
OBS-030	OBS	Info	customer_master (245 files)	245 files = 7 years (2019-2025) x 35 countries, path customer-master/<year>/<country>/. All share an identical 26-column header. 170 files contain non-ASCII text.	Load as UTF-8; capture path metadata.
OBS-031	OBS	High	customer_master	Year folders are yearly snapshots: 260,513 rows for 104,469 unique customers. Customers appear from their acquisition year continuously (no gaps) for 1-7 years. customer_type = NEW only in the first snapshot, RETURNING afterwards.	Grain = (customer_id, snapshot_year).
OBS-032	OBS	Info	customer_master	Attributes that change across snapshots: is_active (7,323 True->False, 9,340 reactivations), loyalty_tier (4,607 moves, upgrades only), postal_code (3,659), phone_number (1,372). Name, DOB, email, segment, country never change.	Model DIM_CUSTOMER as SCD2.
OBS-033	OBS	Info	customer_master	country_code, country_name, region and preferred_language are fully consistent with country_master and the folder. customer_number = CUST-<acq_year>-<CC>-<6 digits>, 1:1 with customer_id.	None.
OBS-034	OBS	Info	customer_master	Segment: Consumer 75%, Business 15%, Education 10%. Loyalty: None 50%, Silver 23%, Gold 16%, Platinum 12%. Gender ~50/50 (Female/Male only).	None.
OBS-035	OBS	Info	customer_master vs sales_header	Customers buy only in their home country (0 mismatches, POS and ONLINE). 85,567 of 104,469 customers have sales; 18,902 (18%) never purchased.	None.
OBS-036	OBS	Medium	customer_master	The snapshot year exists only in the folder path, not in the data.	Derive snapshot_year from METADATA$FILENAME at load.
GAP-030	GAP	High	customer_master	Loading all snapshot files into one table without snapshot_year yields 2.5 rows per customer and inflates customer counts ~2.5x.	Load with snapshot_year; build current view + SCD2.
GAP-031	GAP	High	customer_master vs sales_header	111,385 transactions occur before the customer's registration_date.	Define acquisition date = MIN(first txn, registration).
GAP-032	GAP	Medium	customer_master	6,067 email addresses are shared by 2-10 different customers.	Do not use email as a natural/matching key.
GAP-033	GAP	Medium	customer_master	3,652 snapshot rows have age at registration < 13 (min 10); 22,357 < 18.	Apply a minimum-age rule; COPPA/GDPR review if real.
GAP-034	GAP	Medium	customer_master vs sales_header	40,311 transactions by customers flagged is_active = False in that year's snapshot; 54 in a year the customer is absent from the snapshot.	Clarify the meaning of is_active.
GAP-035	GAP	Medium	customer_master	PII: names, date_of_birth, email, phone, address.	Apply masking / tag-based policies.
GAP-036	GAP	Low	customer_master	Type issues: is_active is the string 'True'/'False'; loyalty_tier uses the string 'None'; created_at mixes 'YYYY-MM-DD' and 'YYYY-MM-DD HH:MI:SS'.	Cast to BOOLEAN / NULL / TIMESTAMP on load.
GAP-037	GAP	Low	customer_master	Trailing whitespace in 2,850 street_address and 16 city values; phone numbers in 8+ formats; 560 rows where state_province = city.	TRIM on load; normalise phone to E.164 if needed.
GAP-038	GAP	Low	customer_master	country_name, region and preferred_language are derivable from country_master.	Drop in curated layer.

Sales

ID	Type	Severity	File(s)	Observation / Gap	Recommendation
OBS-040	OBS	Info	sales_header_*/sales_item_* (14 files)	7 header + 7 item files, 193 MB, identical headers per type. 642,546 headers and 642,546 items; growth ~5.8%/yr (77,155 in 2019 to 107,922 in 2025).	Union into SALES_HEADER / SALES_ITEM.
OBS-041	OBS	Info	sales_header/sales_item	transaction_sk, transaction_id and transaction_line_id are unique. Header:item is exactly 1:1 (line_number always 1). All FKs (customer, store, country, currency, SKU) resolve.	None.
OBS-042	OBS	Info	sales_header/sales_item	All amount arithmetic reconciles to the cent: line_total = qty*unit_price - discount + tax; header totals = sum of lines; net = gross - discount + tax.	Keep as regression checks.
OBS-043	OBS	Info	sales_header	POS (431,617) always has store_id; ONLINE (210,929) never does. The 11 no-store countries have ONLINE sales only (36,690).	Enforce DQ-090.
OBS-044	OBS	Info	sales_header	Currency always equals the country currency; POS country/currency always equal the store's.	None.
OBS-045	OBS	Info	sales_header	Strong Q4 seasonality (Dec 87K vs Feb 37K) with Sep-Oct iPhone launch uplift.	None.
OBS-046	OBS	Info	sales_item	quantity is 1 (79%) or 2 (21%); quantity 2 only for ACC and AIR. category_code always matches the SKU hierarchy.	None.
OBS-047	OBS	Low	sales_header/sales_item	created_at always equals transaction_timestamp; there is no separate ingestion timestamp.	Add load_ts at ingestion.
GAP-040	GAP	Critical	sales_header/sales_item	Amounts are USD-scale but labelled with local currencies (iPhone 11 sells for ~700-1,600 in USD, JPY, KRW, INR alike). 66,607 JPY/KRW transactions have fractional amounts although minor_unit = 0.	Confirm the amount currency; do not FX-convert or sum across currencies until resolved.
GAP-041	GAP	High	sales_header vs tax_master	Tax is added on top in 33 tax-inclusive countries (451,922 transactions). See GAP-007.	Resolve revenue definition.
GAP-042	GAP	High	sales_header vs tax_master	45,671 transactions use a rate different from tax_master (CA, BR, CH, MY, FI); rates are constant over 2019-2025 in sales, so there is no rate history anywhere.	See GAP-001/006.
GAP-043	GAP	High	sales_item vs product_sku_master	6,737 sales occur before the SKU global launch date (6,739 before local launch).	Flag or exclude; fix generator.
GAP-044	GAP	Medium	sales_item	Unit price for the same SKU+currency varies widely: 7,419 of 15,707 SKU/currency pairs have max/min > 2x (max 6.9x). No price list exists.	Introduce a price list for validation.
GAP-045	GAP	Medium	sales_item	Every transaction has a single line, so basket / attach-rate analysis is impossible.	Note limitation.
GAP-046	GAP	Medium	sales_header	Synthetic patterns: 9 payment methods ~11.1% each on both channels, 23,488 ONLINE 'Cash' payments, flat weekday volume, 82,824 POS sales between 00:00 and 05:59.	Do not use for payment-mix or staffing insight.
GAP-047	GAP	Low	sales_header_<year>	161 transactions dated 1 January of the next year sit in the previous year's file; max timestamp 2026-01-01.	Partition by transaction_timestamp, not file name.
GAP-048	GAP	Low	sales_header	Two unique transaction keys (transaction_sk, transaction_id).	Use transaction_sk as PK, transaction_id as alternate key.

Relationship overview